

Graphs & Transformations



Sketching quadratic graphs

- 1 Find the vertex and y -intercept of $y = x^2 - 6x + 5$.
 - 2 Find the vertex and x -intercepts of $y = x^2 - 2x - 8$.
-

Translations of graphs

- 3 The graph of $y = f(x)$ is translated 3 units right and 2 units down. Write the equation of the resulting graph in terms of f .
 - 4 The graph of $y = x^2$ is translated to have vertex $(-2, 5)$. Write its equation.
 - 5 The graph of $y = x^3$ is translated 4 units left and 1 unit up. Write the equation of the resulting graph.
-

Stretches and reflections

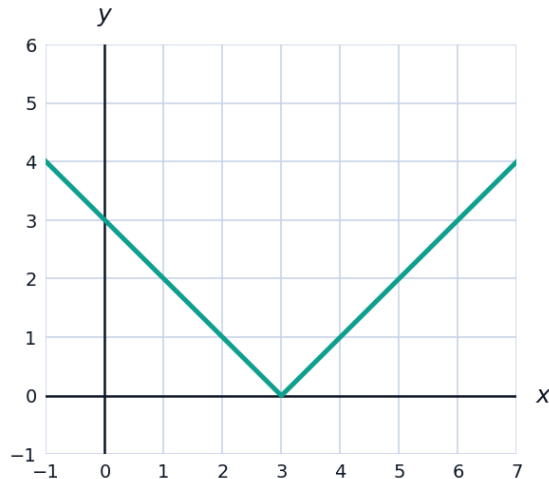
- 6 Describe the transformation that maps $y = x^2$ to $y = 3x^2$.
 - 7 Describe the transformation that maps $y = f(x)$ to $y = f(2x)$.
 - 8 Describe the transformation that maps $y = f(x)$ to $y = -f(x)$.
 - 9 Describe the transformation that maps $y = f(x)$ to $y = f(-x)$.
-

Combining transformations

- 10 Given $f(x) = x^2$, describe the sequence of transformations that produces $g(x) = 2(x - 1)^2 + 3$, in the order they are applied.
 - 11 Given $f(x) = x^2$, describe the sequence of transformations that produces $h(x) = -f(x + 2) + 1$, in the order they are applied.
-

Graphs of the modulus function

- 12 Sketch $y = |x - 3|$ and state its minimum value and where it occurs.



- 13 Solve $|2x - 5| = 7$.
-

Finding the equation of a transformed curve

- 14 A curve $y = f(x)$ has a minimum point at $(4, -2)$. State the coordinates of the minimum point on the graph of $y = f(x - 3) + 5$.
- 15 A curve $y = f(x)$ has a maximum point at $(-1, 6)$. State the coordinates of the maximum point on the graph of $y = 2f(x) - 3$.
-

Intersections of transformed graphs

- 16 Find the coordinates of the points where $y = (x - 2)^2 - 1$ crosses the x-axis.
- 17 Find the coordinates of the points where $y = (x + 1)^2 - 4$ crosses the x-axis.
-

Graphs of reciprocal functions

- 18 State the equations of the asymptotes of $y = \frac{1}{x - 2} + 3$.
- 19 State the equations of the asymptotes of $y = \frac{3}{x + 4} - 2$.
-

A well-designed word problem: modeling a bridge cable

- 20** This question has two parts. A suspension bridge cable hangs in the shape of a parabola. Measuring from the lowest point of the cable, the cable rises to a height of 20 meters at a horizontal distance of 50 meters on each side. (a) Using the lowest point of the cable as the origin, write the equation of the parabola in the form $y = ax^2$, finding a . (b) Find the height of the cable at a horizontal distance of 30 meters from the lowest point.